



2014 Half-Yearly Examination

FORM V

MATHEMATICS 2 UNIT

Wednesday 14th May 2014

General Instructions

- Writing time — 1 hour 30 minutes
- Write using black or blue pen.
- Board-approved calculators and templates may be used.

Total — 80 Marks

- All questions may be attempted.

Section I – 8 Marks

- Questions 1–8 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 72 Marks

- Questions 9–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your name, class and master on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Nine.
- Write your name and master on this question paper and submit it with your answers.

5A: BDD

5B: MLS

5C: LYL

5D: LRP

5E: PKH

5F: BR

5G: SG

5P: REJ

5Q: NL

5R: TCW

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet

Examiner

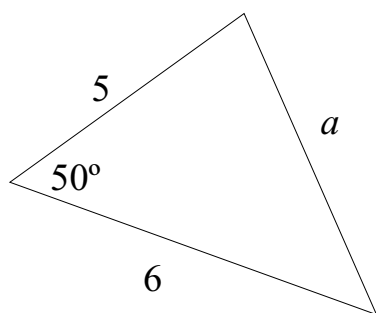
PKH

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Which statement is true for the diagram below?



- (A) $a^2 = 6^2 + 5^2 - 2 \times 6 \times 5 \times \cos 50^\circ$
- (B) $a^2 = 6^2 + 5^2 + 2 \times 6 \times 5 \times \cos 50^\circ$
- (C) $a^2 = 6^2 + 5^2 + 2 \times 6 \times 5 \times \sin 50^\circ$
- (D) $a^2 = 6^2 + 5^2 - 2 \times 6 \times 5 \times \sin 50^\circ$

QUESTION TWO

What is the exact value of $\sqrt{50} - \sqrt{18}$?

- (A) $\sqrt{32}$
- (B) 2.828
- (C) $2\sqrt{2}$
- (D) $2\sqrt{5} - 2\sqrt{3}$

QUESTION THREE

Which of the following is the natural domain of $\sqrt{x-4}$?

- (A) $x \geq 4$
- (B) $x > 4$
- (C) $x \leq 4$
- (D) $x < 4$

QUESTION FOUR

The expression $\frac{K^6 + K^3}{K^3}$ can be simplified to :

- (A) K^6
- (B) $2K^3$
- (C) $K^3 + 1$
- (D) $K^2 + 1$

Exam continues next page ...

QUESTION FIVE

The solution to the equation $\tan \theta = -1$ for $0^\circ \leq \theta \leq 360^\circ$ is :

- (A) $\theta = -45^\circ$ or $\theta = 45^\circ$ (B) $\theta = -45^\circ$
(C) $\theta = 135^\circ$ or $\theta = 225^\circ$ (D) $\theta = 135^\circ$ or $\theta = 315^\circ$

QUESTION SIX

What is the gradient of the line passing through $A(-1, 6)$ and $B(4, -4)$?

- (A) -2 (B) $-\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $-\frac{3}{2}$

QUESTION SEVEN

Which of the following expressions is equal to $a^3 - b^3$?

- (A) $(a - b)(a^2 + b^2)$ (B) $(a + b)(a^2 - ab + b^2)$
(C) $(a - b)(a^2 + 2ab + b^2)$ (D) $(a - b)(a^2 + ab + b^2)$

QUESTION EIGHT

Which of the following is equivalent to $\sin \theta$?

- (A) $\cos(90^\circ + \theta)$ (B) $\cos(90^\circ - \theta)$
(C) $\frac{1}{\sec \theta}$ (D) $-\sin(180^\circ - \theta)$

_____ End of Section I _____

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION NINE (12 marks) Use a separate writing booklet.	Marks
(a) Find $\sqrt{12\,645\,674}$ correct to 3 significant figures.	1
(b) Find θ correct to the nearest degree if $\cos \theta = 0.6$ and θ is acute.	1
(c) Expand $(3x - 2)^2$.	1
(d) Find the mid-point of the interval joining $A(-1, 5)$ and $B(3, -3)$.	1
(e) Evaluate $ 6 - -12 $.	1
(f) Factorise the following:	
(i) $4x^2 - 9$	1
(ii) $2x^2 - 3x + 1$	1
(g) Write $\frac{2x+1}{x^2} + \frac{x-2}{x}$ as a single fraction in simplest form.	1
(h) Solve the following equations:	
(i) $\frac{4x}{3} = x + 1$	1
(ii) $-2(x + 3) = 5(2x - 3)$	2
(i) Solve the inequation $-3a - 2 \leq -8$.	1

QUESTION TEN (12 marks) Use a separate writing booklet. **Marks**

- (a) Find the distance between the points $C(2, -3)$ and $D(-5, 1)$. Leave your answer as a surd. **2**
- (b) Expand and simplify $(2 - 3\sqrt{3})^2$. **2**
- (c) Rationalise the denominator of $\frac{4}{3 - \sqrt{7}}$ and simplify. **2**
- (d) Solve $x = \frac{2 - x}{x}$. **2**
- (e) Solve the quadratic inequation $x^2 - 2x - 8 \leq 0$. **2**
- (f) Solve $|x + 6| = 12$. **2**

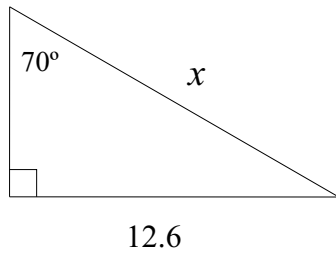
QUESTION ELEVEN (12 marks) Use a separate writing booklet. **Marks**

- (a) If $g(x) = 5x - 3x^2$ find the value of $g(-2)$. **1**
- (b) Sketch neat graphs of the following functions on separate axes showing any intercepts with the axes:
- (i) $y = -\frac{1}{2}x + 3$ **2**
- (ii) $y = 16 - x^2$ **2**
- (iii) $y = \frac{1}{x + 2}$ **2**
- (iv) $y = -\sqrt{9 - x^2}$ **2**
- (c) Write down the natural domain and range of the function $y = 3^x$. **2**
- (d) Sketch the graph of $y = |x - 2|$. **1**

QUESTION TWELVE (12 marks) Use a separate writing booklet.

Marks

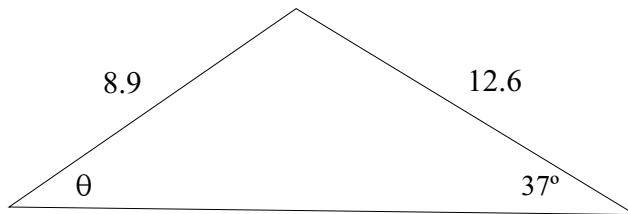
(a)



2

In the diagram above find x correct to two decimal places.

(b)



2

In the diagram above find the acute angle θ correct to the nearest minute.

(c) Find the exact value of $\cos 225^\circ$.

2

(d) Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta \leq 360^\circ$.

2

(e) Draw a neat sketch of $y = \cos x$ for $-180^\circ \leq x \leq 180^\circ$. Label the intercepts with the axes.

2

(f) Find the equation of the line through $P(-2, 4)$ which is perpendicular to $y = \frac{1}{2}x$.

2

QUESTION THIRTEEN (12 marks) Use a separate writing booklet.

Marks

(a) Which of the following numbers are rational?

1

$$\sqrt{7}, \pi, 0.\dot{7}, \sqrt{81}, 7^{\frac{1}{7}}$$

(b) Given that $\cos \theta = -\frac{1}{4}$ and θ is obtuse, find the exact value of $\tan \theta$.

3

(c) Prove that $\frac{\sec \theta \tan \theta}{1 + \tan^2 \theta} = \sin \theta$.

3

(d) Find the perpendicular distance from the point $A(-1, 5)$ to the line $3x - 4y + 2 = 0$.

2

(e) Sketch the region in the number plane which simultaneously satisfies $y \leq 3 - x$ and $y \geq x^2 - 1$. There is no need to find the points of intersection of the graphs.

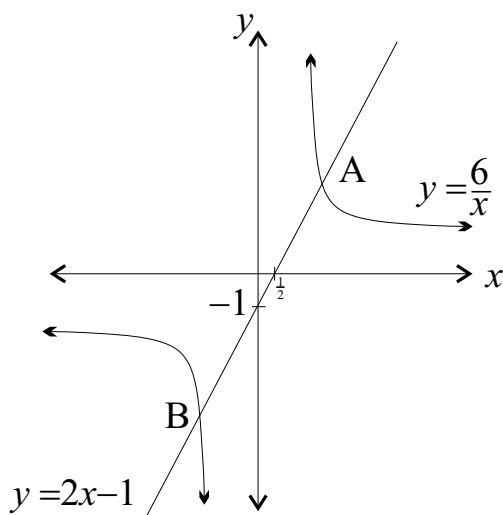
3

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QUESTION FOURTEEN (12 marks) Use a separate writing booklet.

Marks

(a)



In the diagram above the line $y = 2x - 1$ and the hyperbola $y = \frac{6}{x}$ are drawn. The points of intersection of the line and the hyperbola are labelled A and B .

(i) Use algebra to find the co-ordinates of A and B .

3

(ii) Hence solve $2x - 1 \leq \frac{6}{x}$.

2

(b) Two ships leave port P . One travels 90 km on a bearing of 030° T and anchors at A . The other ship travels 40 km on a bearing of 120° T and anchors at B .

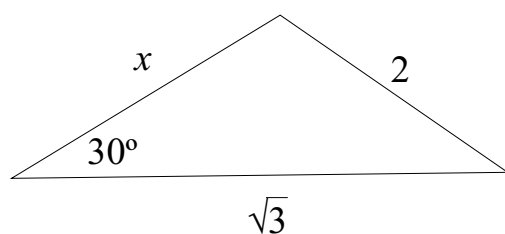
(i) Represent this situation on a neat diagram.

1

(ii) Find the bearing of B from A to the nearest degree.

2

(c)



In the diagram above:

(i) Show that $x^2 - 3x - 1 = 0$.

2

(ii) Find the exact area of the triangle.

2

_____ End of Section II _____

END OF EXAMINATION

Tear-off pages follow ...

B L A N K P A G E



2014

Half-Yearly Examination

FORM V

MATHEMATICS 2 UNIT

Wednesday 14th May 2014

- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

NAME:

CLASS: MASTER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Multiple Choice80

1. A 2. C 3. A 4. C 5. D

8

6. A 7. D 8. B

Q9

(a) $\sqrt{12645674} = 3560$ ✓

(b) $\cos \theta = 0.6$
 $\theta = 53^\circ$ ✓

(c) $(3x-2)^2 = 9x^2 - 12x + 4$ ✓

(d) $M = \left(\frac{3-1}{2}, \frac{5-3}{2} \right) = (1, 1)$ ✓

(e) $|6| - |-12| = 6 - 12 = -6$ ✓

(f) (i) $4x^2 - 9 = (2x-3)(2x+3)$ ✓

(ii) $2x^2 - 3x + 1 = (2x-1)(x-1)$ ✓

(g) $\frac{2x+1}{x^2} + \frac{x^2-2x}{x^2} = \frac{x^2+1}{x^2}$ ✓

(h) (i) $\frac{4x}{3} = x+1$

$4x = 3x + 3$ ✓

$x = 3$

(ii) $-2(x+3) = 5(2x-3)$ ✓

$-2x - 6 = 10x - 15$ ✓

$x = -9$

$x = -\frac{3}{4}$ ✓

(i)

$-3a - 2 \leq -8$

$-3a \leq -6$ ✓

$a \geq 2$

12

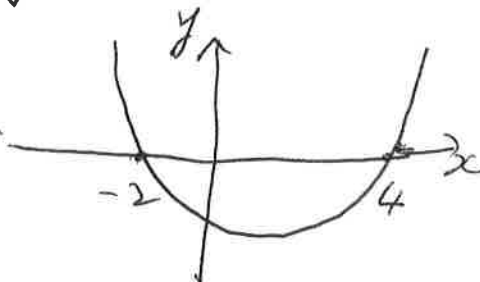
Q10 (a) $d = \sqrt{7^2 + 4^2}$ ✓
 $= \sqrt{65}$ ✓

(b) $(2 - 3\sqrt{3})^2$
 $= 4 - 12\sqrt{3} + 27$ ✓
 $= 31 - 12\sqrt{3}$ ✓

(c) $\frac{4}{3 - \sqrt{7}} \times \frac{3 + \sqrt{7}}{3 + \sqrt{7}}$ ✓
 $= \frac{4(3 + \sqrt{7})}{9 - 7}$ ✓
 $= 6 + 2\sqrt{7}$ ✓

(d) $x = \frac{2 - x}{x}$
 $x^2 + x - 2 = 0$ ✓
 $(x + 2)(x - 1) = 0$
 $x = -2$ or $x = 1$ ✓

(e) $x^2 - 2x - 8 \leq 0$
 $(x + 2)(x - 4) \leq 0$ ✓
 $-2 \leq x \leq 4$ ✓



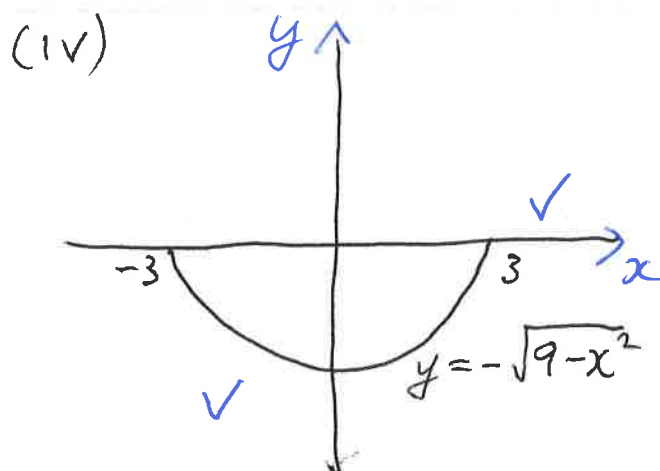
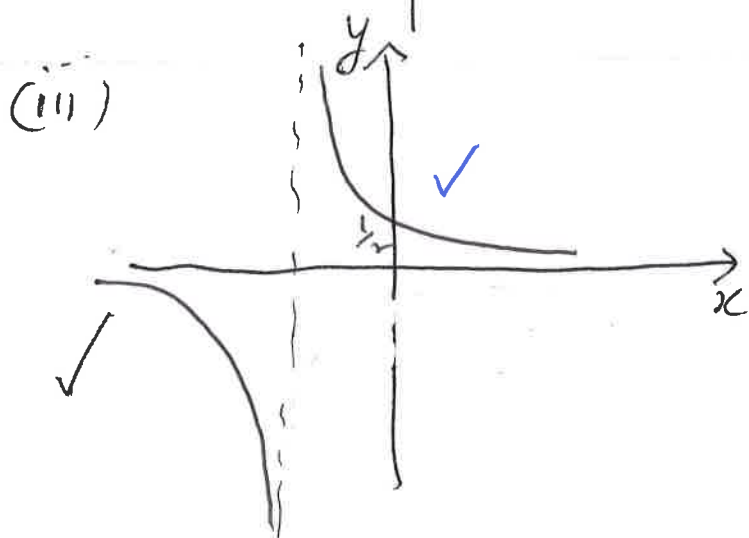
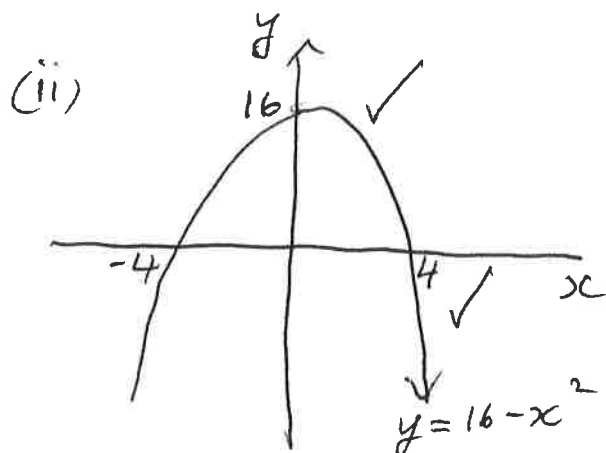
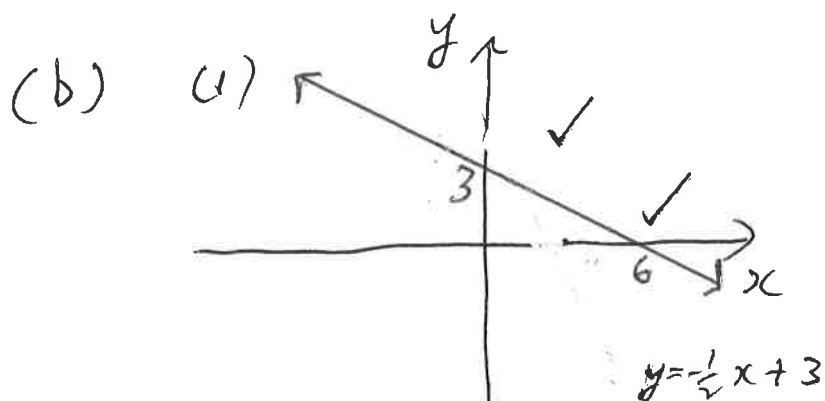
(f) $|x + 6| = 12$

$x + 6 = 12$ or $x + 6 = -12$

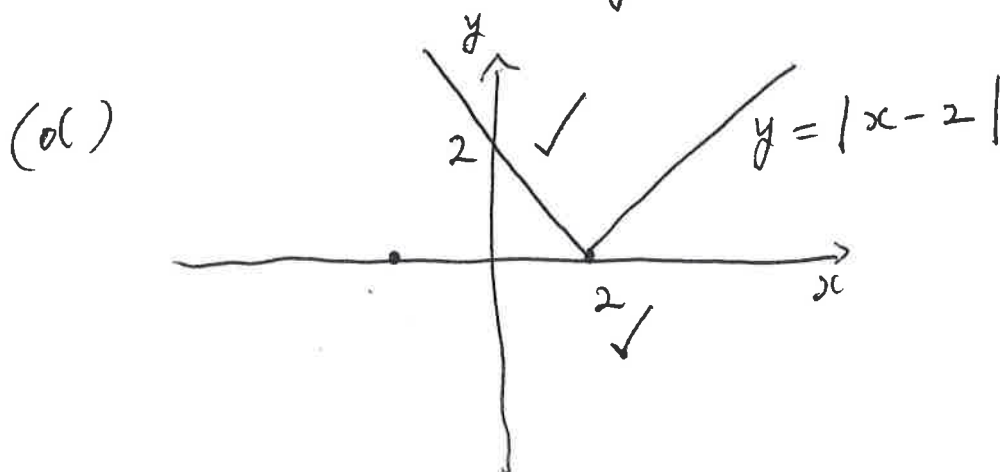
$x = 6$ ✓

$x = -18$ ✓

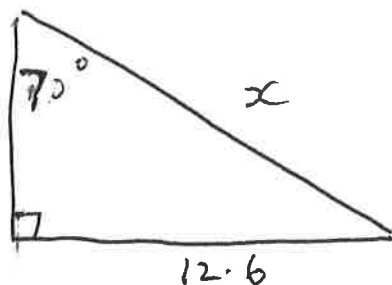
Q 11 (a) $g(-2) = 5(-2) - 3(-2)^2$
 $= -10 - 12$
 $= -22$ ✓



(c) $y = 3^x$ ✓
 Domain : all real x ✓
 Range : $y > 0$ ✓



Q12 (a)



$$\frac{12.6}{x} = \sin 70^\circ \quad \checkmark$$

$$x = 12.6 \div \sin 70^\circ$$

$$x \doteq 13.41 \quad \checkmark$$

$$(b) \quad \frac{\sin \theta}{12.6} = \frac{\sin 37^\circ}{8.9} \quad \checkmark$$

$$\sin \theta = \frac{12.6 \sin 37^\circ}{8.9}$$

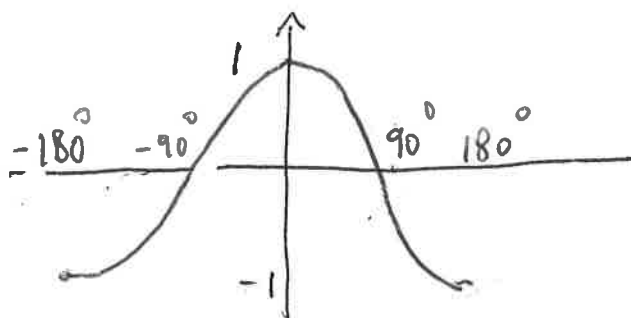
$$\theta \doteq 58^\circ 26' \quad \checkmark$$

$$(c) \quad \cos 225^\circ = -\frac{1}{\sqrt{2}} \quad \checkmark$$

$$(d) \quad \text{Solve } \sin \theta = -\frac{\sqrt{3}}{2} \quad 0^\circ \leq \theta \leq 360^\circ$$

$$\theta = 240^\circ \text{ or } 300^\circ \quad \checkmark \quad \checkmark$$

(e)



(f)

$$y - 4 = -2(x + 2) \quad \checkmark$$

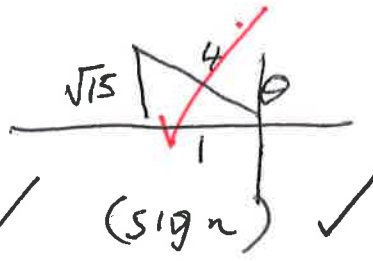
$$y = -2x \quad \checkmark$$

Q13

(a) $0.7, \sqrt{81} \quad \checkmark$

(b) $\cos \theta = -\frac{1}{4}$

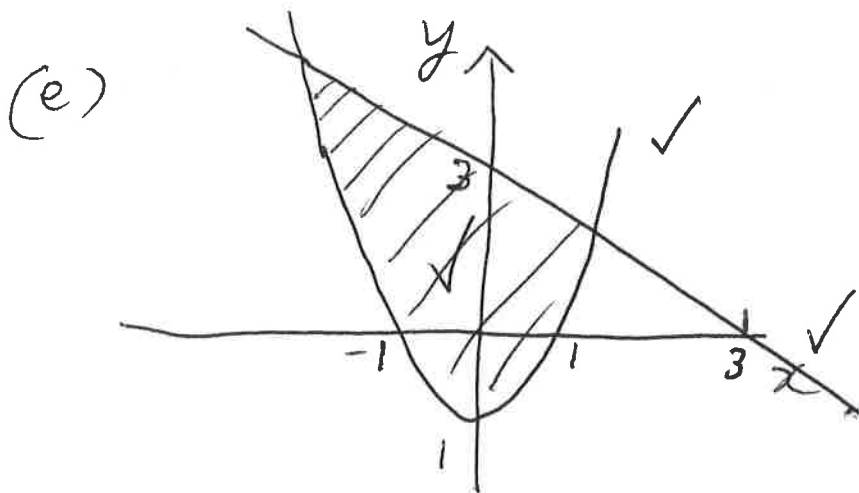
$\tan \theta = -\sqrt{15} \quad \checkmark \quad (\text{sign}) \quad \checkmark$



(c)
$$\begin{aligned} \text{LHS} &= \frac{\sec \theta \tan \theta}{1 + \tan^2 \theta} = \frac{\sec \theta \tan \theta}{\sec^2 \theta} \quad \checkmark \\ &= \frac{\tan \theta}{\sec \theta} \quad \checkmark \\ &= \frac{\sin \theta}{\cos \theta} \times \cos \theta \quad \checkmark \\ &= \sin \theta = \text{RHS} \end{aligned}$$

(d) $A(-1, 5) \quad 3x - 4y + 2 = 0$
 $(x_1, y_1) \quad a \quad b \quad c$

$$\begin{aligned} d &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \quad \checkmark \\ &= \frac{|-3 - 20 + 2|}{5} = \frac{21}{5} \text{ units} \quad \checkmark \end{aligned}$$

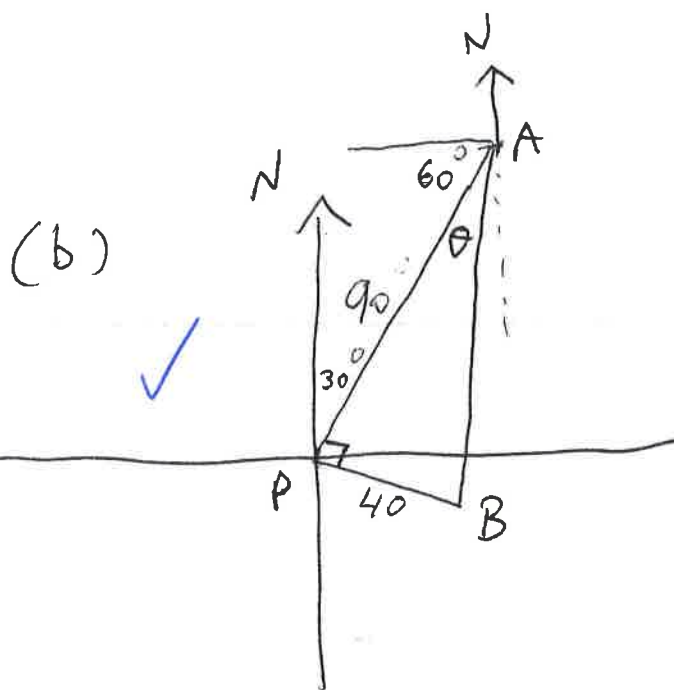


Q14 (a) (i) Pts of $\cap$ where $2x-1 = \frac{6}{x}$ ✓

(ii) Solution is ✓✓

$x \leq -\frac{3}{2}$ or $0 < x \leq 2$

$$\begin{aligned} 2x^2 - x &= 6 \\ 2x^2 - x - 6 &= 0 \\ (2x+3)(x-2) &= 0 \quad \checkmark \\ x = -\frac{3}{2} \quad \text{or} \quad x = 2 \quad \checkmark \\ y = -4 \quad \checkmark \quad y = 3 \quad \checkmark \end{aligned}$$



Now $\tan \theta = \frac{4}{9}$

$\theta = 24^\circ$ ✓

Bearing B from A is 186° T. ✓

(c) By the cosine rule,

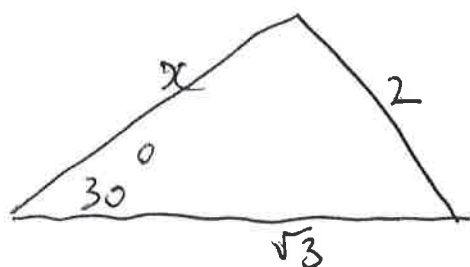
(i)

$$4 = x^2 + 3 - 2\sqrt{3}x \cos 30^\circ \quad \checkmark$$

$$4 = x^2 + 3 - 2\sqrt{3}x \cdot \frac{\sqrt{3}}{2}$$

$$4 = x^2 + 3 - 3x$$

$$x^2 - 3x - 1 = 0 \quad \checkmark$$



(ii)

$$x = \frac{3 \pm \sqrt{9+4}}{2} \quad \checkmark$$

$$x = \frac{3 + \sqrt{13}}{2}$$

$$\text{Area} = \frac{1}{2} a b \sin 30^\circ$$

$$= \frac{1}{2} \sqrt{3} \left(\frac{3 + \sqrt{13}}{2} \right) \cdot \frac{1}{2}$$

$$= \frac{3\sqrt{3} + \sqrt{39}}{8} \quad \checkmark$$